House Price Prediction Using Linear Regression

Introduction

The objective of this project is to build and evaluate a Linear Regression model for predicting house prices using the California Housing Dataset. This project demonstrates the complete machine learning workflow including data loading, exploratory data analysis (EDA), model training, prediction, and performance evaluation.

Dataset Description

The California Housing Dataset is a popular dataset available in Scikit-Learn. It contains information collected from California districts, including features related to housing and population statistics.

Features Used

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

Target Variable

- Median House Value (MedHouseVal)

The dataset contains 20,640 records and 8 input features.

Exploratory Data Analysis (EDA)

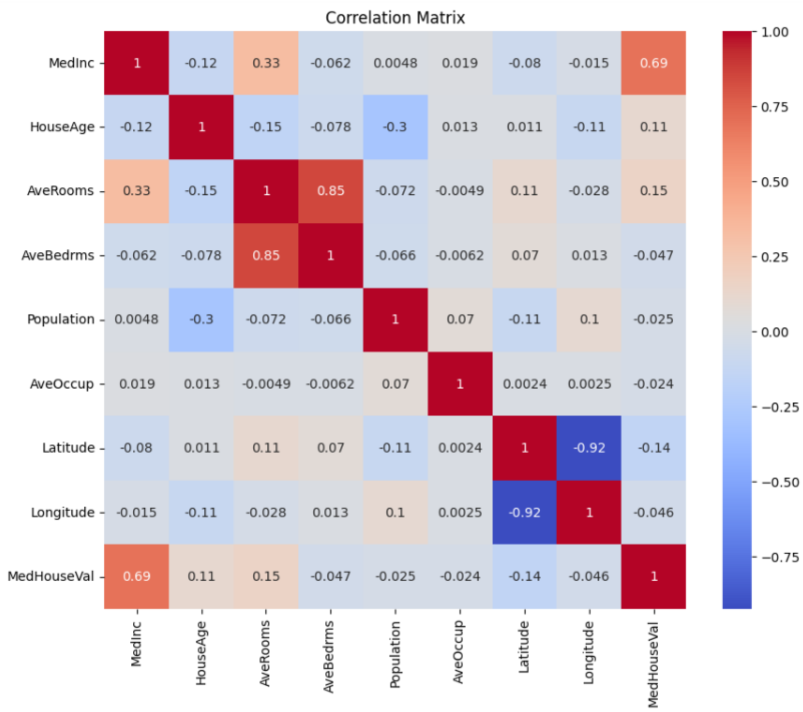
The dataset was analyzed to understand its structure and relationships between variables.

Data Quality Check

- No missing values were found in the dataset.
- Summary statistics were generated for all features.
- Feature distributions and correlations were examined.

Correlation Analysis

A correlation heatmap was created to visualize relationships between variables.



Key Findings

- Median Income (MedInc) showed the strongest positive correlation with house prices (0.69).
- Latitude and Longitude exhibited strong negative correlation.
- Other features showed moderate to weak relationships with house prices.

Model Development

Algorithm Used

Linear Regression was selected because it is one of the most widely used supervised machine learning algorithms for predicting continuous numerical values.

Data Splitting

The dataset was divided into:

- Training Set: 80%

- Testing Set: 20%

The model was trained using the training data and evaluated using unseen test data.

Model Evaluation

The model performance was measured using three evaluation metrics:

Mean Absolute Error (MAE)

MAE = 0.5332

Root Mean Squared Error (RMSE)

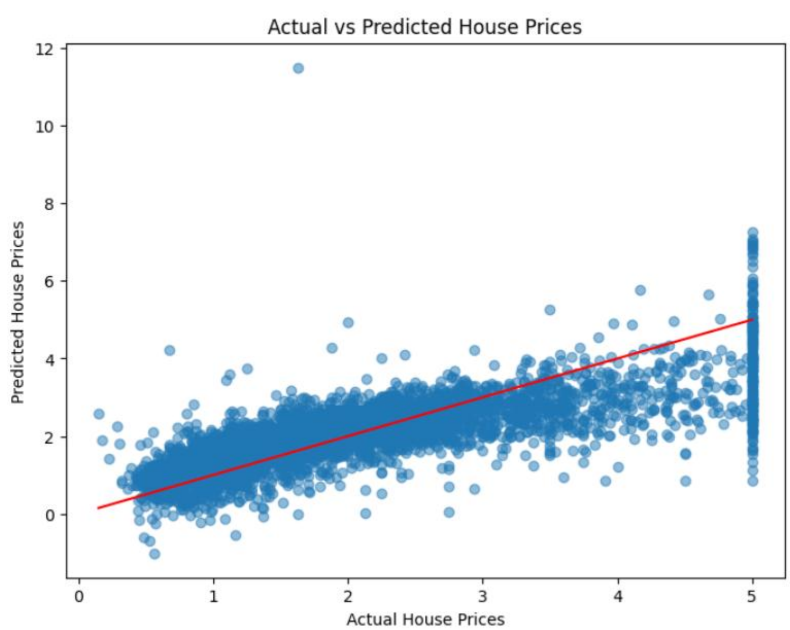
RMSE = 0.7456

R^2 Score

R^2 Score = 0.5758

These metrics indicate that the model provides a reasonable prediction of housing prices and explains approximately 57.58% of the variance in the target variable.

Prediction Visualization



The scatter plot demonstrates the relationship between actual and predicted house prices. Most predictions follow the overall trend of the actual values, indicating that the model has learned meaningful patterns from the dataset.

Conclusion

In this project, a Linear Regression model was successfully developed to predict housing prices using the California Housing Dataset. The project covered all major stages of the machine learning pipeline, including data exploration, preprocessing, model training, prediction, and evaluation.

The model achieved:

- MAE: 0.5332
- RMSE: 0.7456
- R^2 Score: 0.5758

The results demonstrate moderate predictive performance and provide a strong foundation for understanding regression-based machine learning models.

Future Improvements

Future work may include:

- Feature Engineering
- Data Normalization
- Hyperparameter Optimization
- Random Forest Regression
- Gradient Boosting Models
- XGBoost and Ensemble Methods

These techniques could further improve prediction accuracy and model performance.